BUSINESS REPORT

Topic: Pattern Discovery with Unsupervised Learning

- ANIRUDHA K R

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1. Context and Objective

Context:

AllLife Bank wants to focus on its credit card customer base in the next financial year. They have been advised by their marketing research team, that the penetration in the market can be improved. Based on this input, the Marketing team proposes to run personalized campaigns to target new customers as well as upsell to existing customers. Another insight from the market research was that the customers perceive the support services of the back poorly. Based on this, the Operations team wants to upgrade the service delivery model, to ensure that customer queries are resolved faster. The Head of Marketing and Head of Delivery both decide to reach out to the Data Science team for help

Objective:

To identify different segments in the existing customers, based on their spending patterns as well as past interaction with the bank, using clustering algorithms, and provide recommendations to the bank on how to better market to and service these customers.

2. *Exploratory Data Analysis*

❖ Problem Definition:

- The bank currently treats its credit card customers as a single customer base despite differences in spending capacity and interaction preferences. This limits the effectiveness of marketing campaigns and results in inefficient customer support allocation.
- Using historical customer data including credit limits, number of credit cards, bank visits, online visits, and customer support calls the bank seeks to identify meaningful customer segments through clustering techniques.

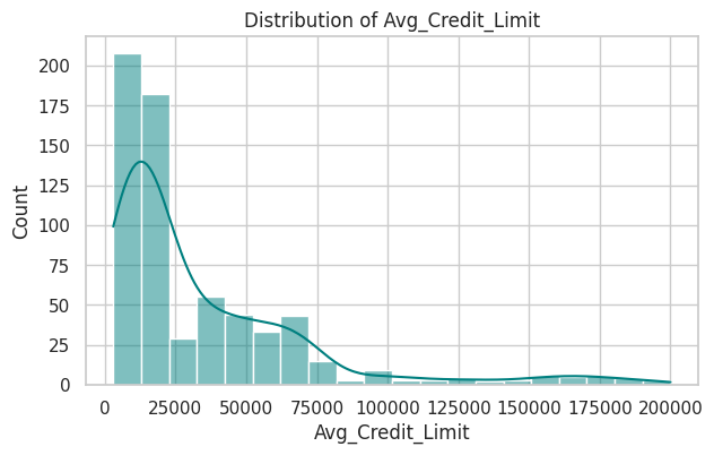
  AllLife Bank wants to understand customer behavior based on:
  - Credit Limit
  - Number of Credit Cards
  - Bank Visits
  - Online Visits
  - Customer Service Calls

- The objective is to identify distinct customer groups that can be targeted through personalized marketing strategies and improved customer service.

  These segments will help the bank:
  - Improve personalized marketing campaigns.
  - Increase credit card usage and cross-selling opportunities.
  - Enhance customer satisfaction through customized service models.
  - Allocate operational resources more efficiently.

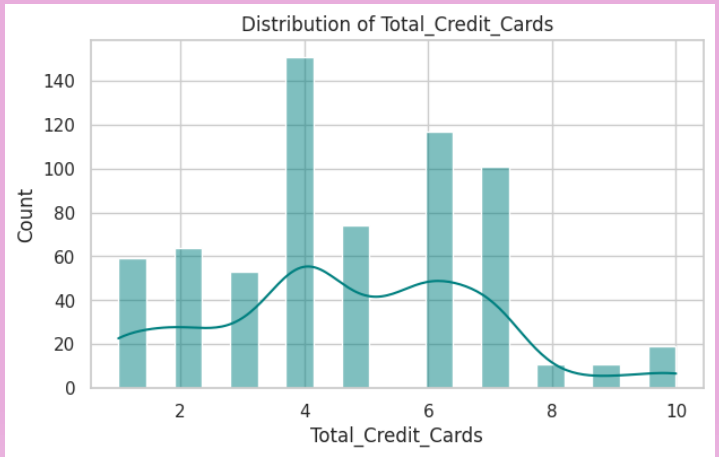
❖ Univariate analysis:

➢ Numerical Variables



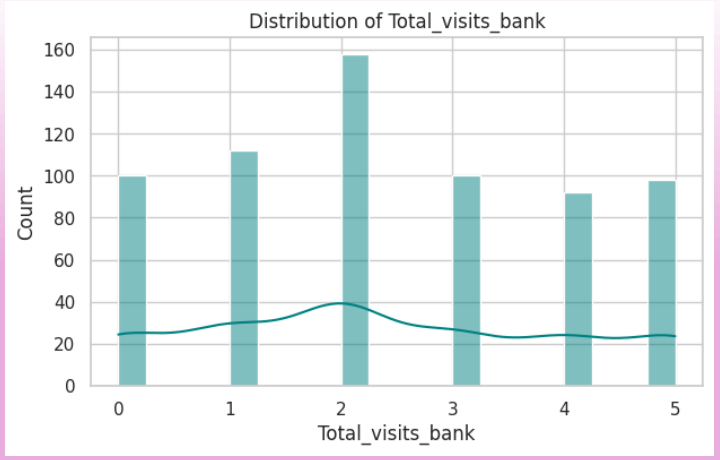
1. Average Credit Limit

- The distribution is right-skewed, indicating that while most customers have moderate credit limits, a small proportion enjoy substantially higher limits.
- Several high-value outliers suggest the presence of premium or affluent customers.
- This variable is expected to be a key factor in distinguishing customer segments.



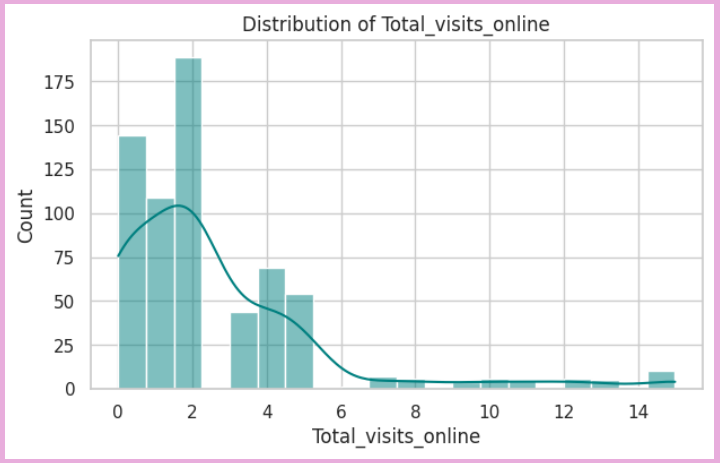
2. Total Credit Cards

- Most customers own 2–6 credit cards, while relatively few own a very high number.
- The distribution indicates opportunities for cross-selling additional credit card products to customers with fewer cards.
- Customers with multiple cards are likely to have stronger relationships with the bank.



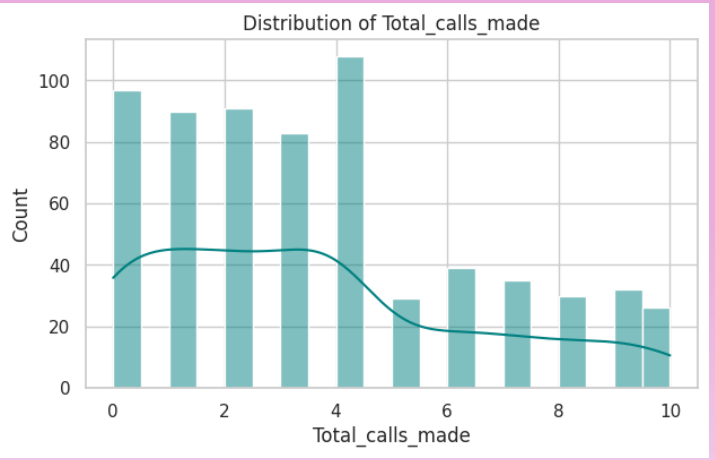
3. Total Bank Visits

- The majority of customers make few branch visits, suggesting limited dependence on physical banking services.
- A smaller group visits branches frequently, indicating a preference for traditional banking channels.



4. Total Online Visits

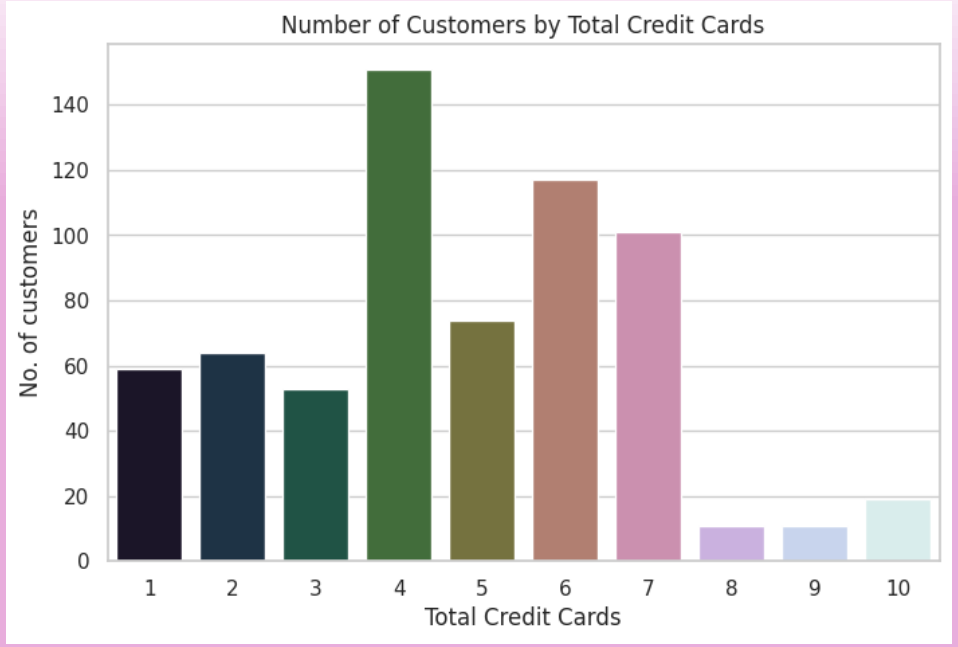
- Online banking usage varies considerably across customers.
- A substantial proportion of customers actively use digital banking, reflecting increasing adoption of online services.
- Some customers rarely use online banking, highlighting an opportunity to promote digital channels.



5. Total Calls Made

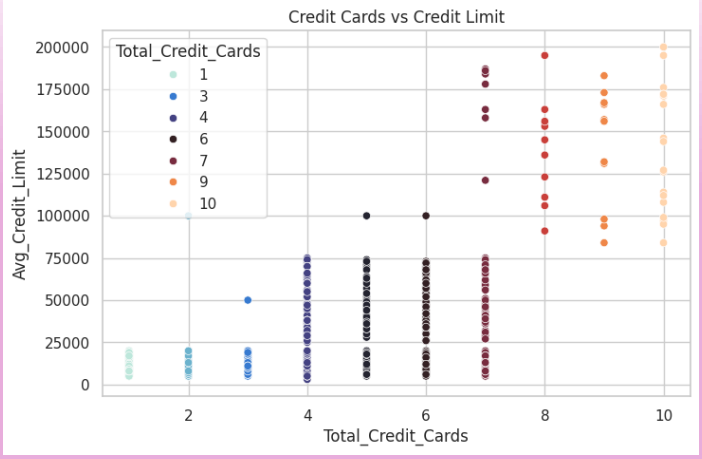
- Most customers make only a small number of customer service calls.
- A few customers contact support frequently, which may indicate service issues, higher engagement, or greater assistance requirements.
- This variable can help identify customers who require improved service experiences.

➢ Count Plot



- Most customers own 4 credit cards, making it the largest customer group (approximately 150 customers). This indicates that four credit cards is the most common level of credit card ownership among customers.
- Customers with 6 and 7 credit cards also form a significant portion of the customer base, suggesting that many customers maintain multiple credit card relationships with the bank.
- The number of customers with 1, 2, 3, and 5 credit cards is comparatively lower and fairly similar, indicating moderate representation in these categories.
- Very few customers own 8, 9, or 10 credit cards, making these high-card-ownership customers a niche segment. They may represent premium, loyal, or high-income customers.
- The distribution is not uniform; it is concentrated around 4 to 7 credit cards, with customer frequency declining sharply beyond 7 cards.
- The presence of customers with a high number of credit cards suggests an opportunity for the bank to identify and reward high-value customers with premium products and exclusive benefits.
- Customers with 1 to 3 credit cards present an opportunity for cross-selling additional credit cards based on their spending behavior, creditworthiness, and financial needs.

❖ Bivariate Analysis:

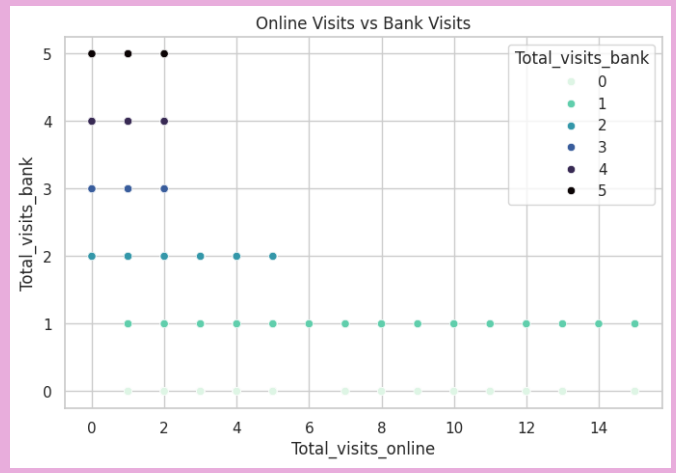


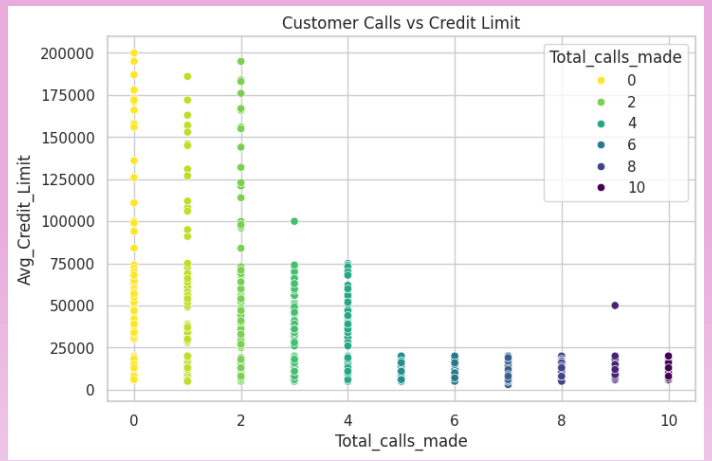
1. Average Credit Limit vs Total Credit Cards

- Customers with higher credit limits generally own more credit cards.
- This suggests that financially valuable customers are more likely to adopt multiple banking products.

2. Online Visits vs Bank Visits

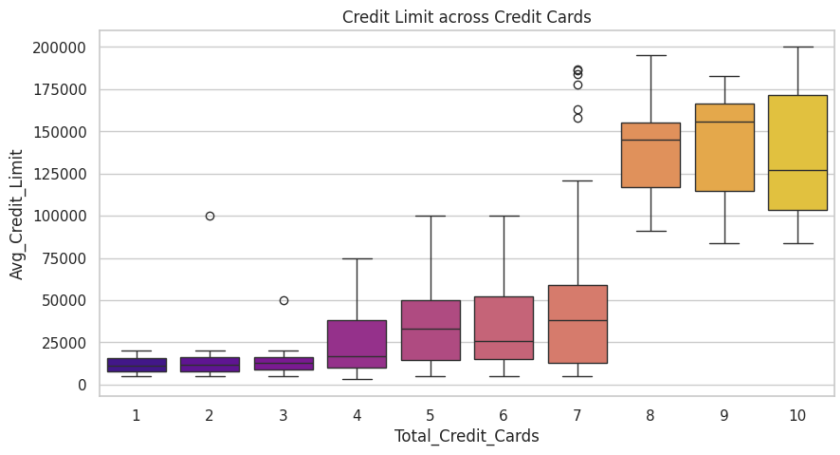
- An inverse relationship is observed between online visits and branch visits.
- Customers who actively use online banking tend to visit physical branches less frequently.
- This indicates a shift toward digital banking among certain customer groups.





3. Average Credit Limit vs Customer Service Calls

- There is only a weak relationship between credit limit and customer service calls.
- Service usage appears to depend more on customer behavior than spending capacity.

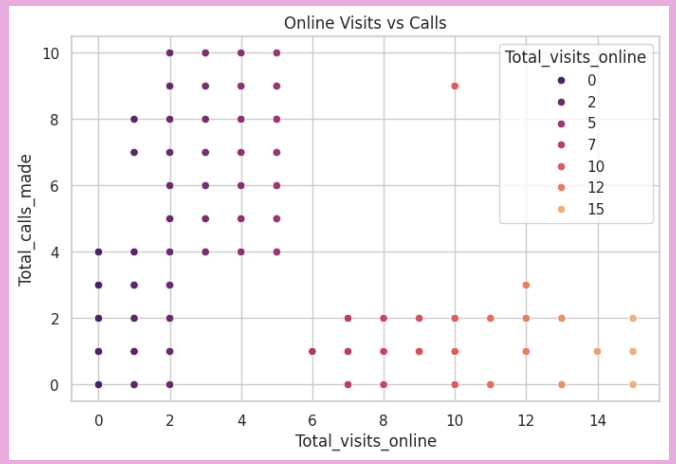


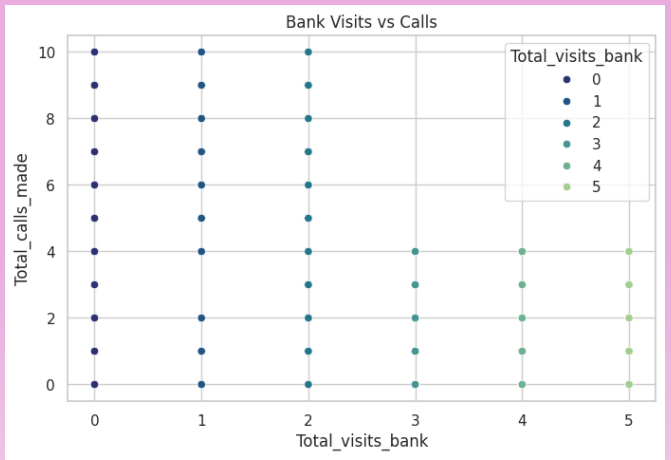
4. Average Credit Limit vs Total Credit Cards

- A positive relationship exists between the number of credit cards and the average credit limit.
- Customers owning more credit cards generally have higher average credit limits, indicating that the bank extends greater credit to customers with stronger financial profiles.
- Customers with only one or two credit cards mostly have lower credit limits, suggesting potential opportunities for future credit limit enhancements.

5. Online Visits vs Customer Service Calls

- There is no strong linear relationship between the number of online visits and customer service calls.
- Customers with high online activity generally tend to make fewer customer service calls, suggesting that digital channels help customers resolve many of their needs independently.
- A small group of customers exhibits both high online visits and high call volumes, which may indicate complex service requirements or unresolved issues.

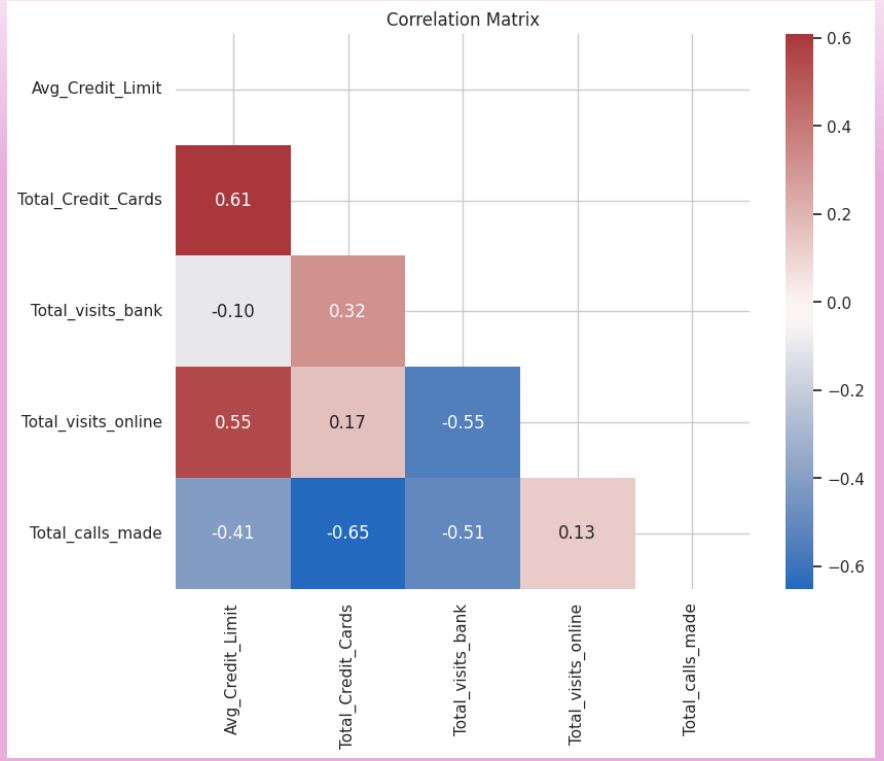




6. Bank Visits vs Customer Service Calls

- Customers who frequently visit branches often make more customer service calls.
- These customers may prefer traditional service channels or require additional assistance.

❖ Correlation Matrix:



- High-value customers are more digitally engaged and require less customer support, making them ideal candidates for premium products and digital-first marketing campaigns.
- Traditional customers rely more on branch visits, while digitally active customers prefer online banking, highlighting the need for channel-specific engagement strategies.
- The strong negative relationship between credit card ownership and customer service calls suggests that loyal, experienced customers are generally more self-sufficient.
- The weak correlations among several variables indicate that each feature captures a different aspect of customer behavior, supporting the use of clustering techniques to identify distinct customer segments for personalized marketing and service improvement.

❖ Key meaningful observations

- **Avg Credit Limit**
  - Right-skewed distribution.
  - Few customers have exceptionally high credit limits.
  - Indicates the presence of premium customers.
- **Total Credit Cards**
  - Most customers own only a few credit cards.
  - A smaller segment owns multiple cards, suggesting cross-selling potential.
- **Bank Visits**
  - Majority of customers visit bank branches infrequently.
  - A subset relies heavily on branch services.
- **Online Visits**
  - Many customers actively use online banking.
  - Digital engagement varies significantly across customers.
- **Customer Service Calls**
  - Most customers make relatively few service calls.
  - Customers with frequent calls may indicate dissatisfaction or require additional support.
- **Correlation Insights**
  - Average Credit Limit tends to increase with the number of credit cards, indicating higher-value customers often maintain multiple cards.
  - Online Visits and Bank Visits generally exhibit an inverse relationship, suggesting that digitally engaged customers depend less on physical branches.
  - Customer Service Calls may show weak relationships with other variables, indicating service interactions are driven by individual needs rather than spending behavior.

❑ Overall EDA Conclusion

The EDA suggests substantial variation in customer financial behavior and service preferences. Differences in credit limits, card ownership, digital usage, and support interactions indicate that the customer base is heterogeneous. These patterns support the use of clustering algorithms (such as K-Means and Hierarchical Clustering) to identify meaningful customer segments for targeted marketing campaigns and tailored service strategies.

3. *Data Preprocessing*

➢ Dataset Overview

- **Shape:** In the dataset there are 660 rows and 5 columns
- **Information:**

```
 Column                 Non-Null Count  Dtype
---------------------------------------------------------------------------
 0   Avg_Credit_Limit   660 non-null    int64
 1   Total_Credit_Cards 660 non-null    int64
 2   Total_visits_bank  660 non-null    int64
 3   Total_visits_online 660 non-null   int64
 4   Total_calls_made   660 non-null    int64
dtypes: int64(5)
```

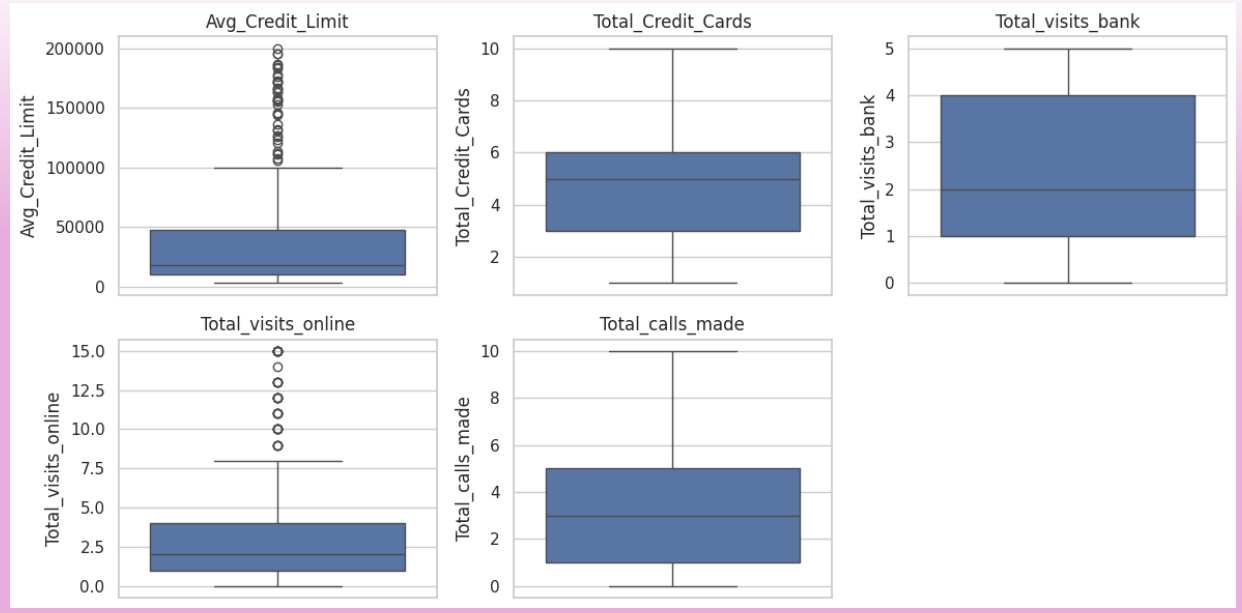
- **Description:**

	count	mean	std	min	25%	50%	75%	max
Avg_Credit_Limit	660.0	34574.242424	37625.487804	3000.0	10000.0	18000.0	48000.0	200000.0
Total_Credit_Cards	660.0	4.706061	2.167835	1.0	3.0	5.0	6.0	10.0
Total_visits_bank	660.0	2.403030	1.631813	0.0	1.0	2.0	4.0	5.0
Total_visits_online	660.0	2.606061	2.935724	0.0	1.0	2.0	4.0	15.0
Total_calls_made	660.0	3.583333	2.865317	0.0	1.0	3.0	5.0	10.0

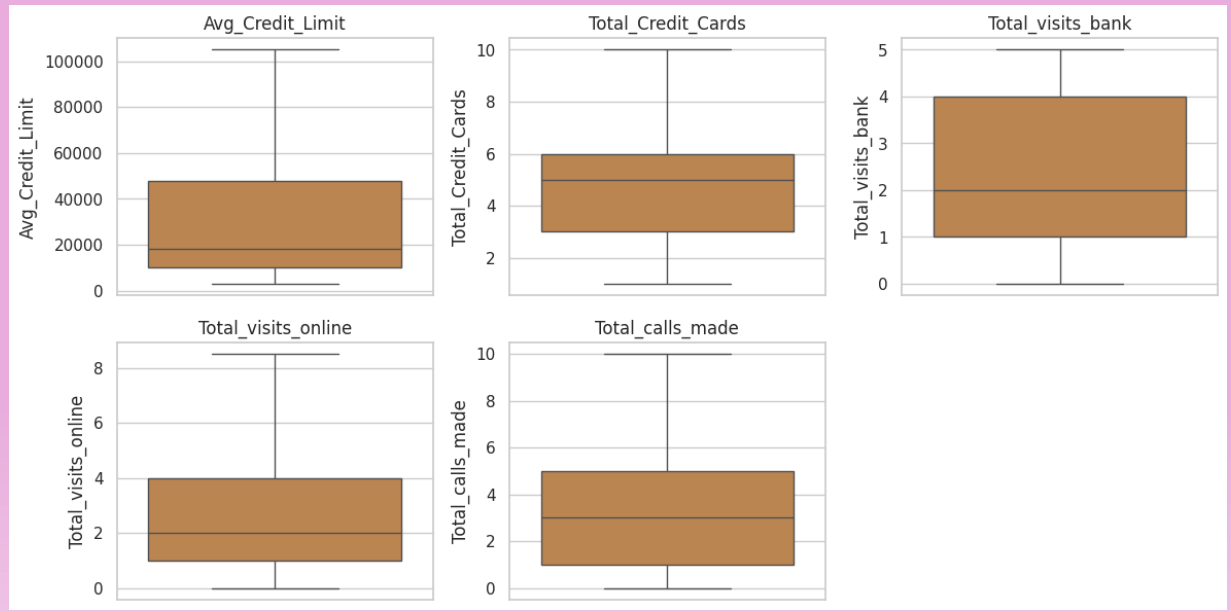
- **Missing Value Treatment**: There are no missing values in the dataset
- **Duplicates**: There are no duplicates in the dataset

➢ Outlier Detection:

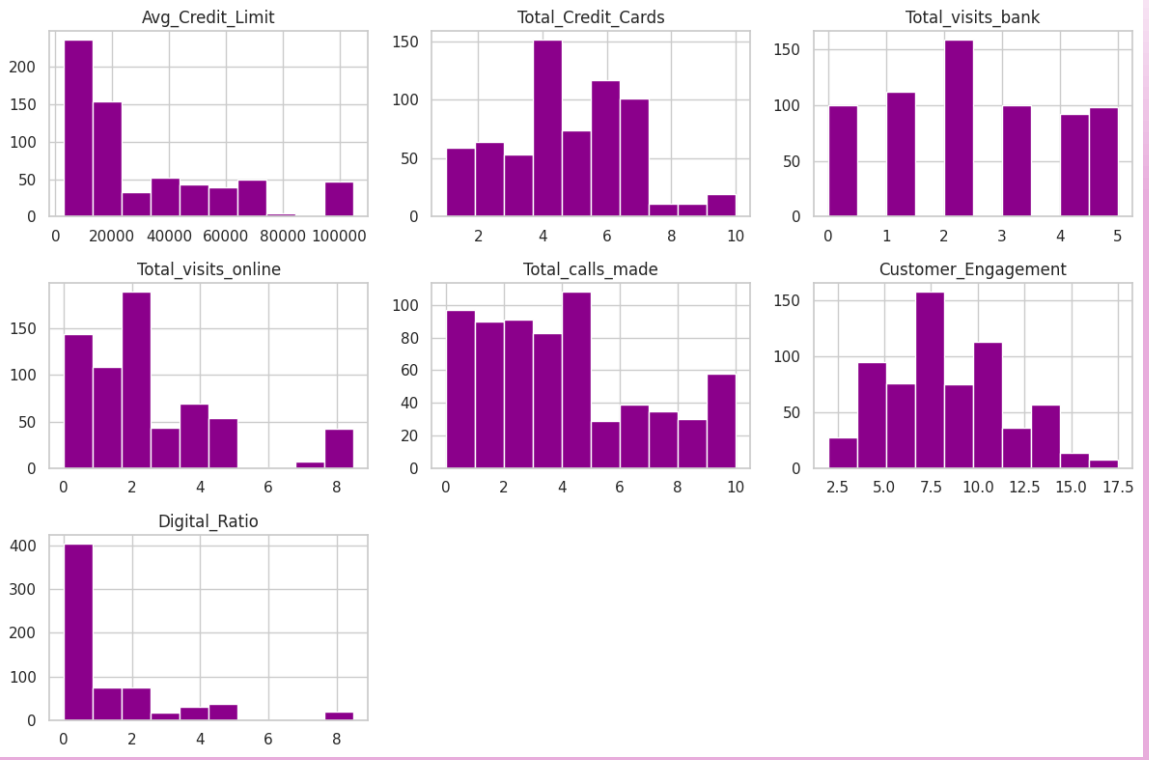
Avg_Credit_Limit :	39
Total_Credit_Cards :	0
Total_visits_bank :	0
Total_visits_online :	37
Total_calls_made :	0



➢ Outlier Treatment:



➢ Feature Engineering



- The engineered features Customer_Engagement and Digital_Ratio effectively summarize customer interaction behavior.
- Customer_Engagement identifies customers with high and low levels of interaction, supporting customer retention and engagement strategies.
- Digital_Ratio distinguishes digitally active customers from branch-dependent customers, enabling targeted digital transformation initiatives.
- Together with the original variables, these engineered features provide a richer understanding of customer behavior and improve the interpretability of customer segments during clustering analysis.

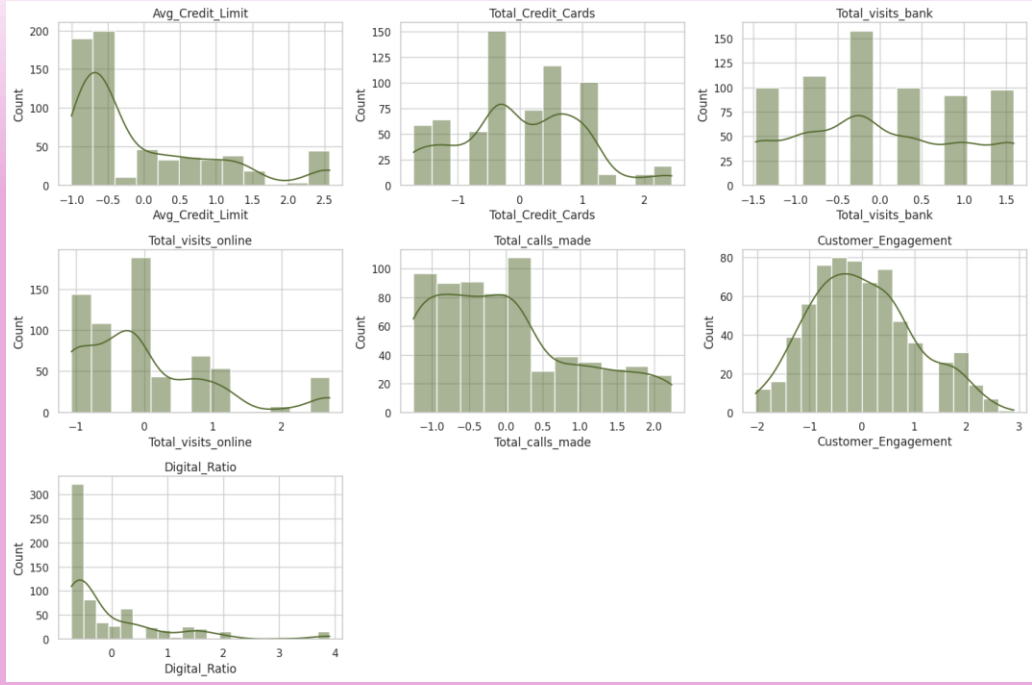
➢ Data Scaling

- Since K-Means uses Euclidean distance, variables with larger magnitudes dominate the clustering process.
  For example:
  - Avg Credit Limit ranges from thousands to hundreds of thousands.
  - Visits range from 0–15.
- Without scaling, Credit Limit would dominate all other variables.

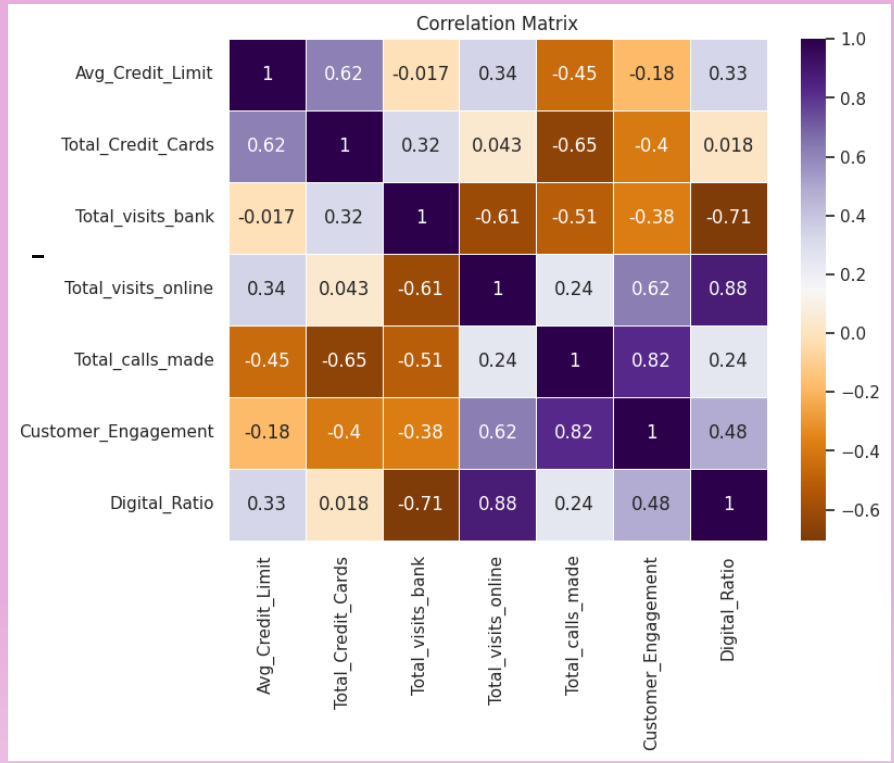
• Description after scaled:

	count	mean	std	min	25%	50%	75%	max
Avg_Credit_Limit	660.0	4.306320e-17	1.000758	-1.006385	-0.760640	-0.479788	0.573406	2.574474
Total_Credit_Cards	660.0	4.306320e-17	1.000758	-1.710864	-0.787585	0.135694	0.597334	2.443892
Total_visits_bank	660.0	-4.306320e-17	1.000758	-1.473731	-0.860451	-0.247170	0.979390	1.592670
Total_visits_online	660.0	-9.689219e-17	1.000758	-1.063178	-0.619767	-0.176357	0.710465	2.705813
Total_calls_made	660.0	-8.612639e-17	1.000758	-1.251537	-0.902271	-0.203739	0.494794	2.241125
Customer_Engagement	660.0	-2.153160e-16	1.000758	-2.032908	-0.759174	-0.122307	0.514560	2.902811
Digital_Ratio	660.0	4.306320e-17	1.000758	-0.721771	-0.617419	-0.449550	0.367113	3.905986

- **Distribution After Scaling**



- **Correlation After Scaling**



- ## Final Dataset Ready for Clustering

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Customer_Engagement	Digital_Ratio
0	2.398942	-1.249225	-0.860451	-0.619767	-1.251537	-2.032908	-0.449550
1	0.643619	-0.787585	-1.473731	2.705813	1.891859	2.902811	3.905986
2	0.643619	1.058973	-0.860451	0.267054	0.145528	-0.122307	0.094892
3	-0.058511	0.135694	-0.860451	-0.619767	0.145528	-0.759174	-0.449550
4	2.398942	0.597334	-1.473731	2.705813	-0.203739	0.992210	3.905986

- ## Key Observations

Missing Values

- No missing values were found in the dataset (or they were handled using median imputation).
- The dataset is complete and suitable for clustering.

Duplicate Records

- Duplicate customer records were removed (if any) to prevent biased cluster formation.

Identifier Variables

- Sl_No and Customer Key were removed because they uniquely identify customers but do not reflect customer behavior.

Outlier Detection

- Outliers were identified using the IQR method.
- Extreme values mainly occurred in Avg_Credit_Limit, representing premium customers.
- Winsorization (capping) was applied instead of deletion to preserve valuable customer information while limiting undue influence on distance calculations.

Feature Engineering

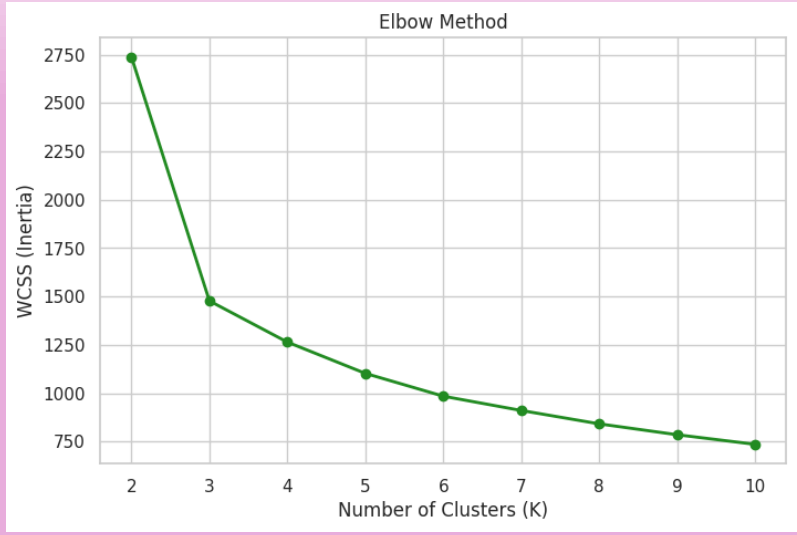
- Optional behavioral features such as Customer_Engagement and Digital_Ratio were created to capture overall interaction and digital preference.
- These features can improve cluster interpretability if included.

Data Scaling

- Standardization (StandardScaler) was applied to ensure all features contribute equally to the Euclidean distance used by clustering algorithms.
- This prevents variables with larger numerical ranges (such as Avg_Credit_Limit) from dominating the clustering process.

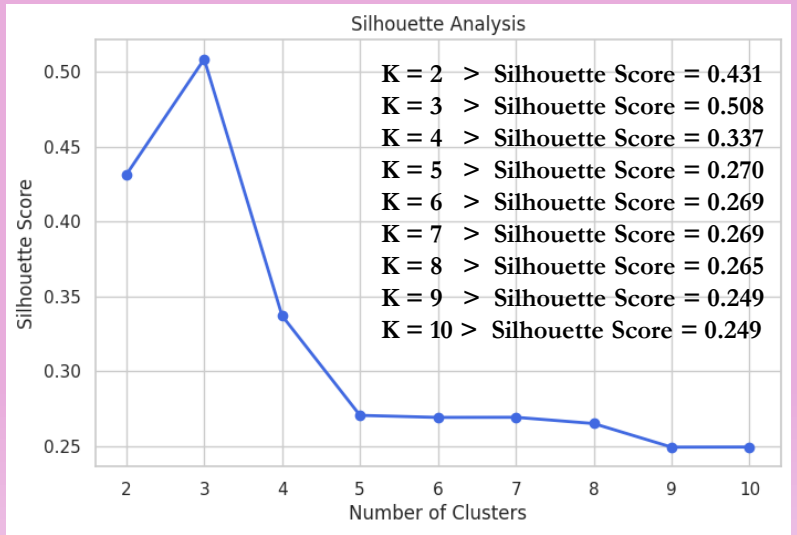
4. *K-means Clustering*

➢ Elbow Method



- The Within-Cluster Sum of Squares (WCSS) decreases sharply as the number of clusters increases from K = 2 to K = 3, indicating that adding a third cluster significantly improves cluster compactness.
- After K = 3, the rate of decrease in WCSS becomes much more gradual, showing diminishing returns from adding additional clusters.
- Three clusters are sufficient to capture the major differences in customer behavior, including spending patterns and banking interactions.
- Choosing more than three clusters would increase model complexity while providing only marginal improvements in clustering performance.

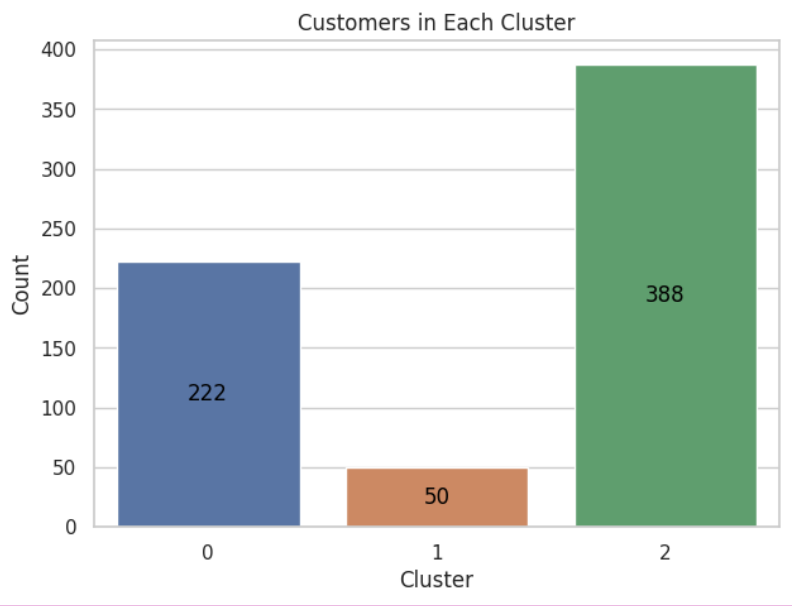
➢ Silhouette Scores



- The highest Silhouette Score is achieved when K = 3, indicating the best cluster separation and cohesion among all the tested values.
- Although K = 2 also produces a reasonably high Silhouette Score, it is lower than that of K = 3, suggesting that two clusters do not capture the customer diversity as effectively.
- The Silhouette Score drops sharply after K = 3, indicating that increasing the number of clusters results in poorer cluster quality and greater overlap between customer groups.
- A Silhouette Score of approximately 0.51 indicates good clustering quality, meaning customers within each cluster are similar to one another while remaining well separated from customers in other clusters.
- Using three clusters enables the bank to design targeted marketing campaigns, tailor customer service strategies, and optimize product offerings without introducing unnecessary complexity.

➢ Train the Final K-Means Model

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Customer_Engagement	Digital_Ratio	Cluster
0	100000.0	2.0	1.0	1.0	0.0	2.0	0.5	2
1	50000.0	3.0	0.0	8.5	9.0	17.5	8.5	0
2	50000.0	7.0	1.0	3.0	4.0	8.0	1.5	2
3	30000.0	5.0	1.0	1.0	4.0	6.0	0.5	2
4	100000.0	6.0	0.0	8.5	3.0	11.5	8.5	1
5	20000.0	3.0	0.0	1.0	8.0	9.0	1.0	0
6	100000.0	5.0	0.0	8.5	2.0	10.5	8.5	1
7	15000.0	3.0	0.0	1.0	1.0	2.0	1.0	2
8	5000.0	2.0	0.0	2.0	2.0	4.0	2.0	0
9	3000.0	4.0	0.0	1.0	7.0	8.0	1.0	0



Cluster 0 – High-Value Customers (33.6%)

- Highest average credit limit.
- Own multiple credit cards.
- Frequently use online banking.
- Rarely visit branches or call customer service.

Cluster 1 – Traditional Banking Customers (7.6%)

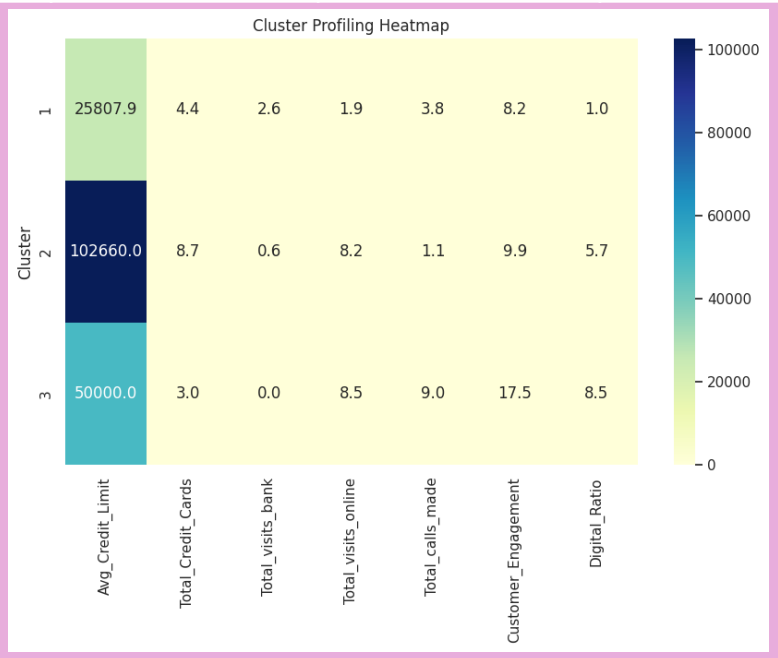
- Moderate credit limits.
- Prefer visiting bank branches.
- Lower online engagement.
- Moderate number of credit cards.

Cluster 2 – Service Dependent Customers (58.8%)

- Lower credit limits.
- Fewer credit cards.
- Make more customer service calls.
- Lower digital engagement.

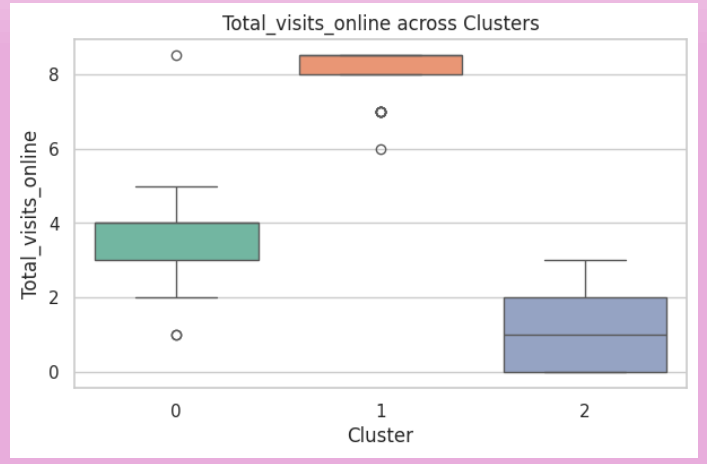
➢ Cluster Profiling

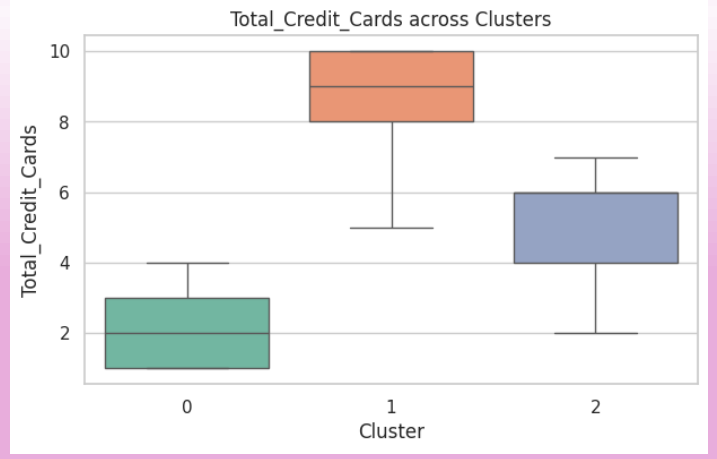
Cluster	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Customer_Engagement	Digital_Ratio
0	12184.684685	2.400901	0.932432	3.565315	6.909910	11.407658	2.234985
1	102660.000000	8.740000	0.600000	8.180000	1.080000	9.860000	5.745000
2	33664.948454	5.505155	3.476804	0.984536	2.002577	6.463918	0.235954

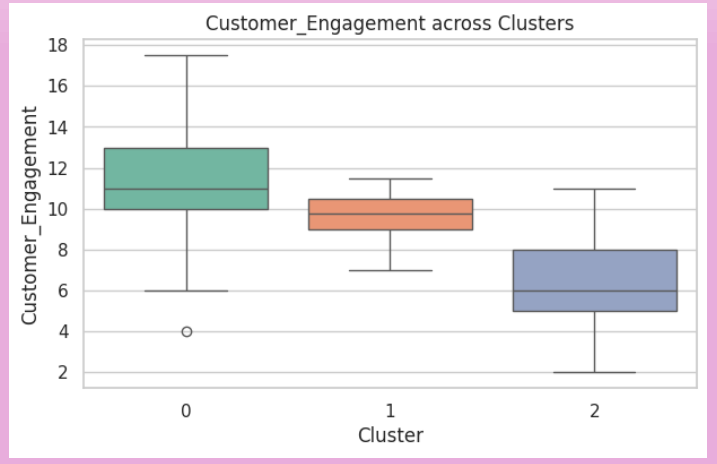


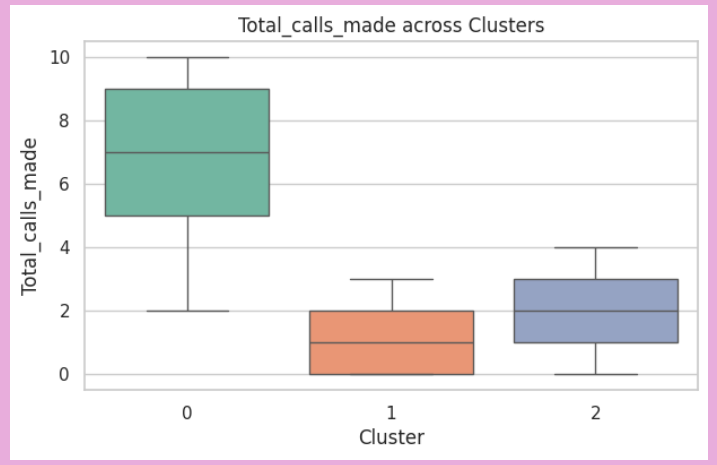
- Customers are highly engaged, interacting frequently through online banking and customer support.
- Despite high digital usage, they make the highest number of customer service calls, suggesting unresolved issues or greater service needs.
- They own fewer credit cards and have only moderate credit limits, indicating opportunities for cross-selling.
- Cluster 1 consists of moderate-value customers with low engagement and limited digital adoption. They represent an opportunity for cross-selling and digital banking promotion.
- Cluster 2 comprises premium, high-value customers who are digitally active, require minimal support, and should be retained through exclusive benefits and personalized services.
- Cluster 3 includes highly engaged digital customers who rely heavily on customer support. Improving service quality and proactively resolving issues can significantly enhance their experience while creating opportunities to increase product adoption.

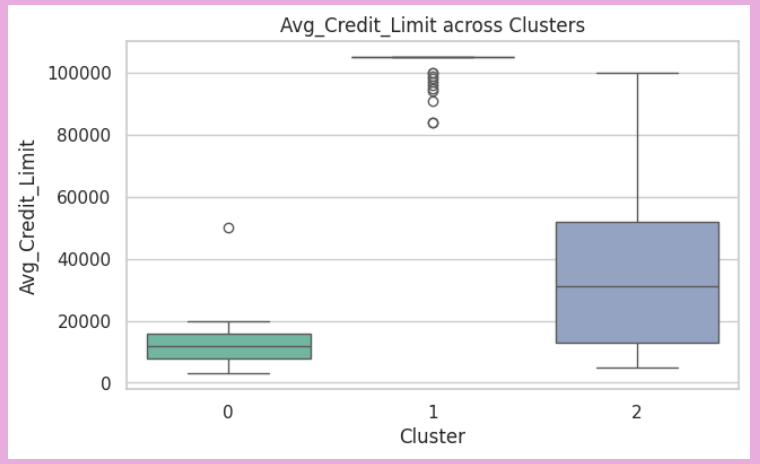
➢ Boxplot of Clusters Profiling

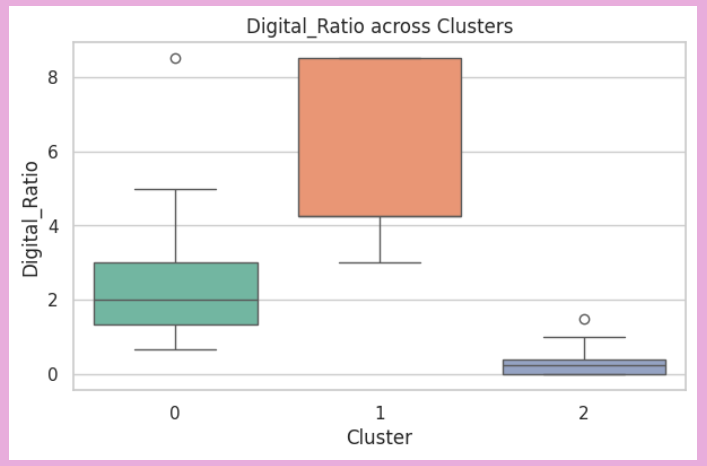


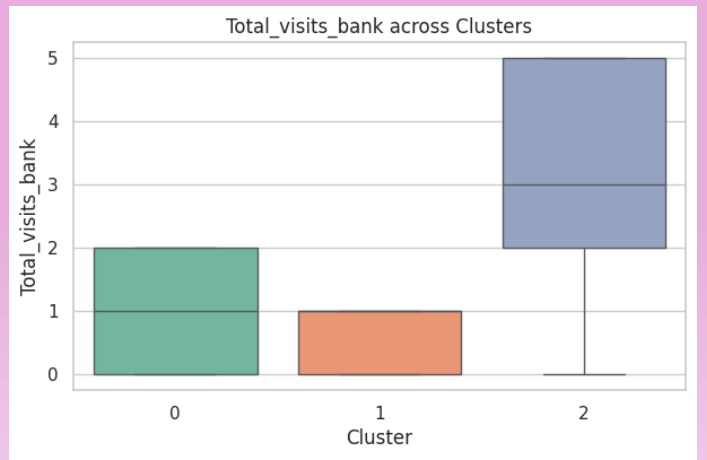




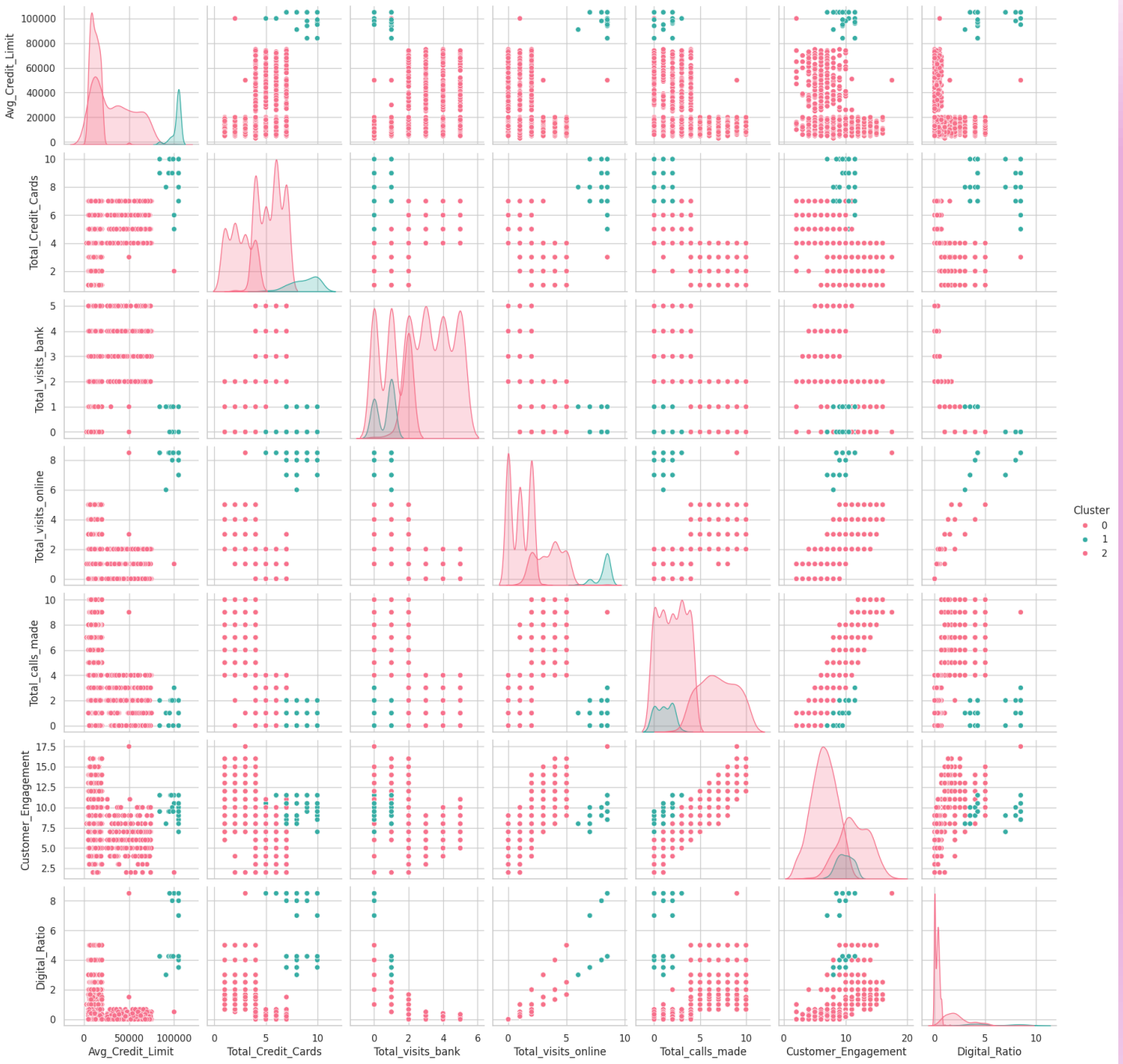






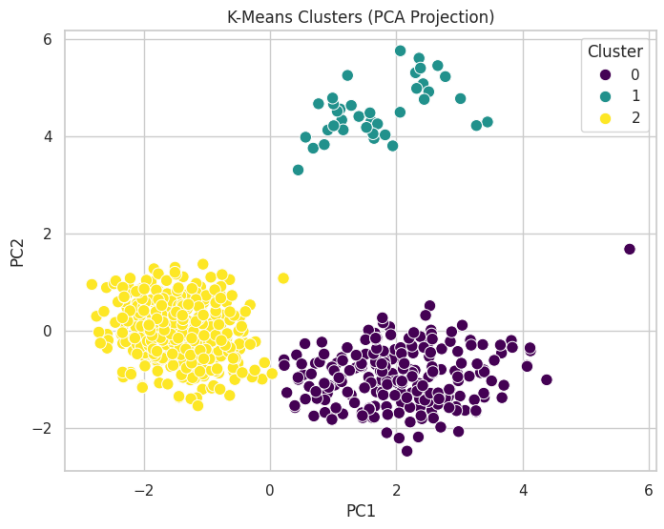


➢ Pairplot of Clusters Profiling



- Cluster 1 consists of high-value customers with the highest credit limits and the most credit cards.
- Cluster 2 includes customers with high online visits and high customer engagement, but they also make more customer service calls.
- Cluster 0 contains customers with moderate credit limits, fewer online visits, and average engagement.
- Customers with higher credit limits generally own more credit cards.
- Online visits increase customer engagement, while customers with more bank visits tend to have lower digital ratios.
- The clusters are well separated, indicating that the K-Means model has effectively identified distinct customer segments based on their banking behavior.

➢ PCA Visualization of Clusters Profiling



- The PCA plot clearly shows three distinct customer clusters, indicating successful segmentation by the K-Means algorithm.
- Cluster 1 is well separated from the other clusters, representing customers with unique characteristics.
- Clusters 0 and 2 are also clearly distinguishable with very little overlap.
- Most customers are concentrated in Clusters 0 and 2, while Cluster 1 contains fewer customers.
- The clear separation between clusters suggests that the selected K = 3 provides meaningful and well-defined customer segments.

➢ Cluster Centroids

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Customer_Engagement	Digital_Ratio
0	-0.683944	-1.064153	-0.901888	0.517721	1.161861	0.962805	0.495049
1	2.492325	1.862226	-1.105763	2.563922	-0.874330	0.469979	2.406049
2	0.070153	0.368893	0.658524	-0.626624	-0.552105	-0.611448	-0.593308

➢ Final Conclusion

Based on the Elbow Method and Silhouette Analysis, the customer base can be effectively segmented into three clusters. The cluster profiles reveal distinct differences in credit usage, interaction channels, and service needs. These insights enable AllLife Bank to design personalized marketing campaigns, enhance customer experience, and optimize operational resource allocation.

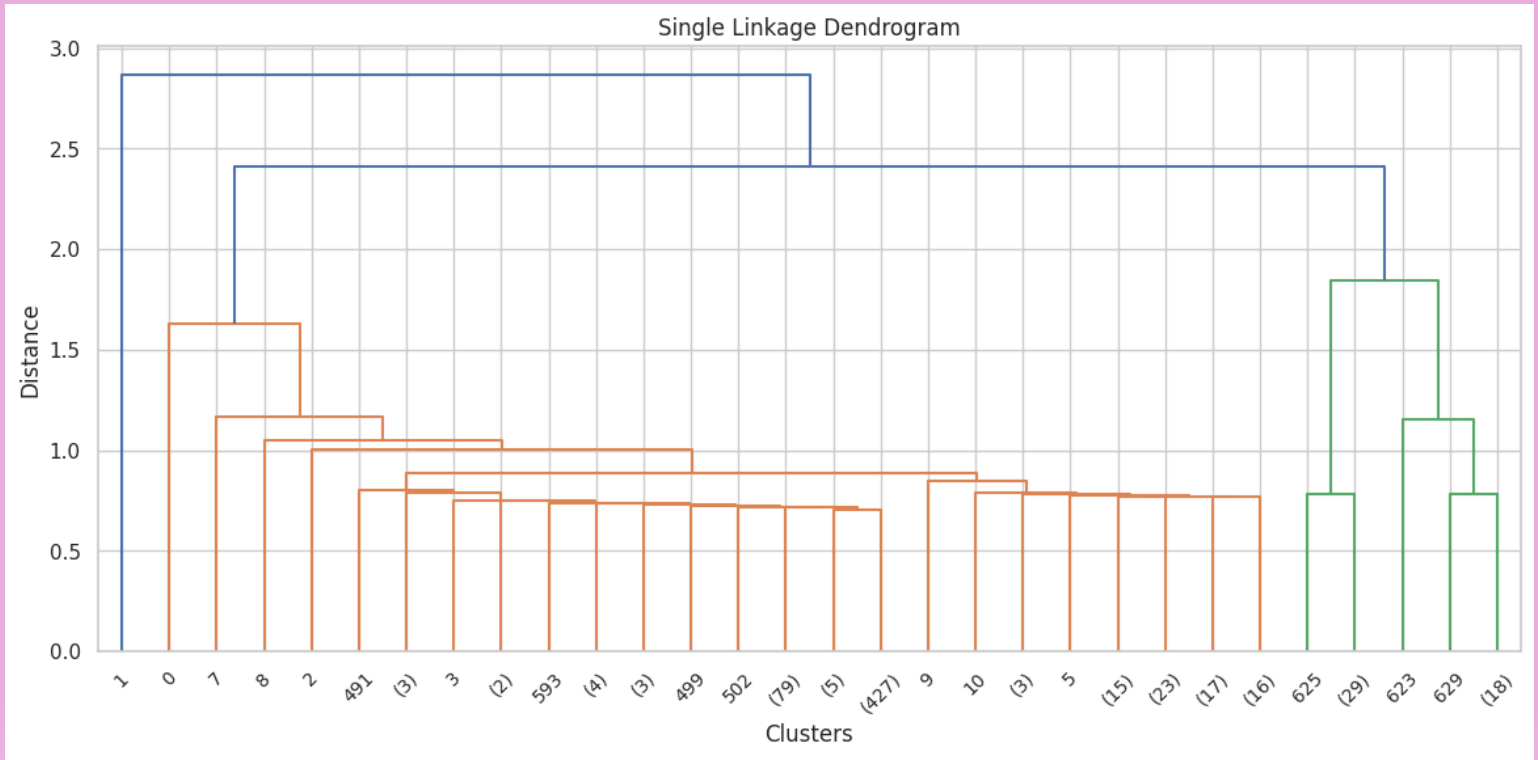
5. *Hierarchical Clustering*

➢ Linkage Methods

We'll compare four linkage methods:

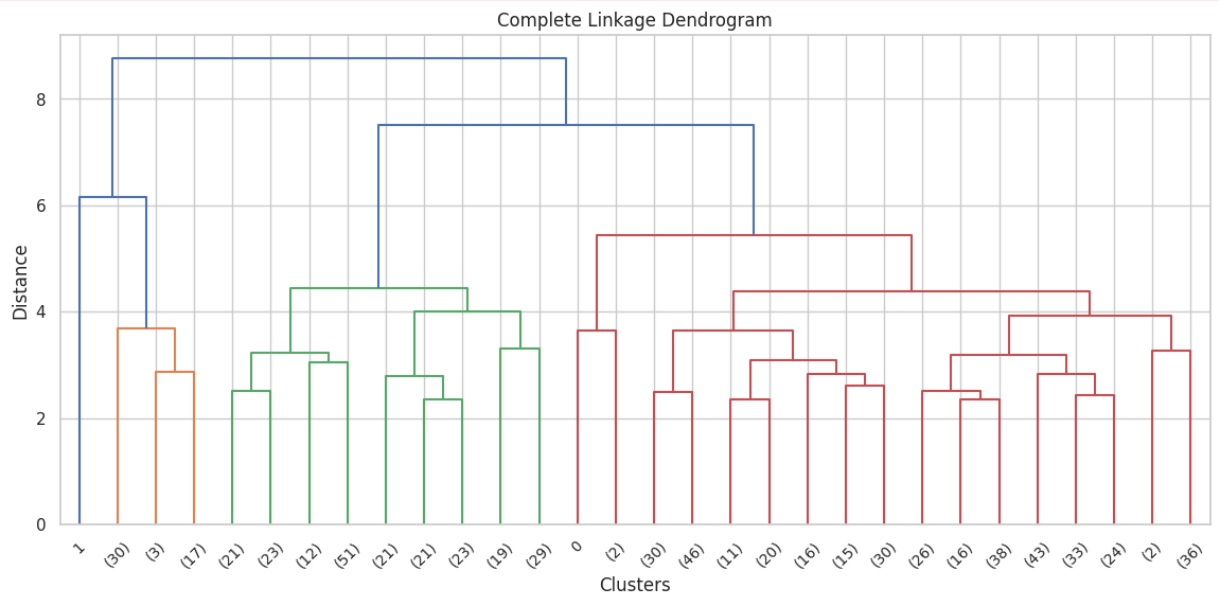
- Single
- Complete
- Average
- Ward

a. Single Linkage



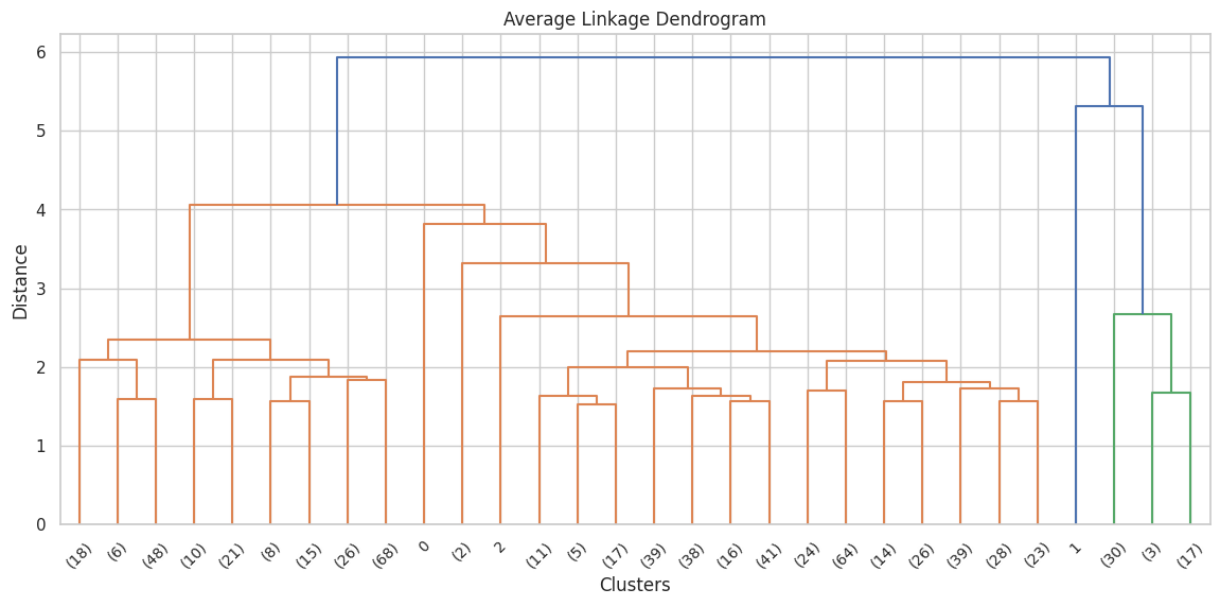
- The dendrogram shows that customers are grouped step by step based on their similarity.
- Most customers merge at low distances, indicating they are quite similar.
- A few customers merge only at higher distances, suggesting they are different from the majority.
- The clusters are not very well separated due to the chaining effect of single linkage.
- Single linkage is less suitable for this dataset compared to Complete, Average, or Ward linkage methods.

b. Complete Linkage



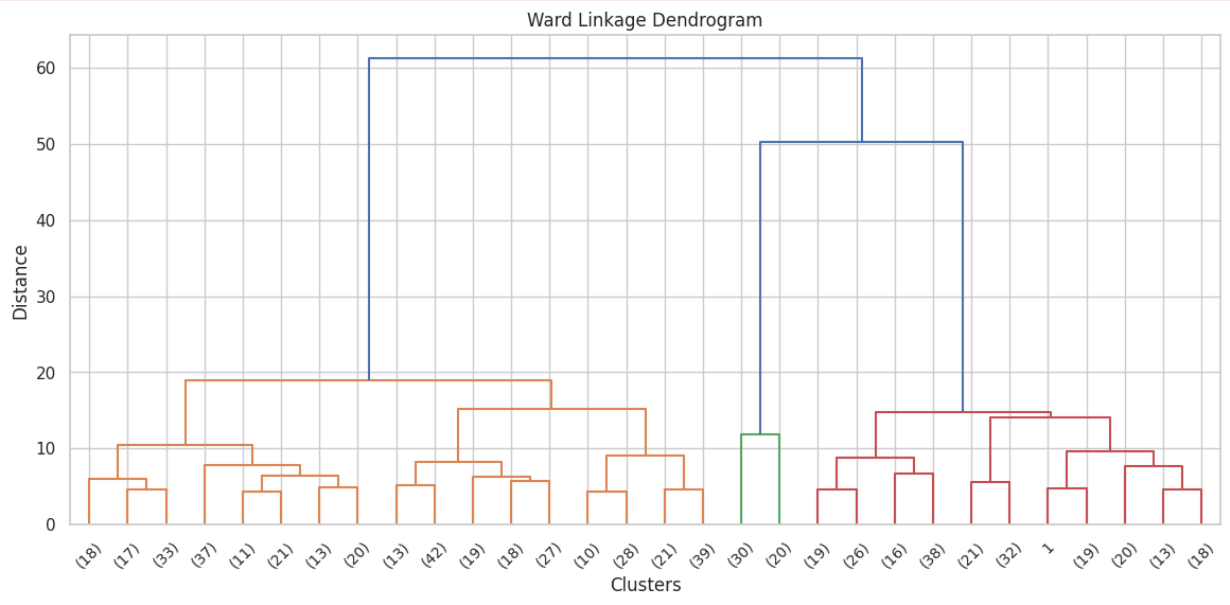
- The dendrogram forms clear and well-separated clusters compared to Single Linkage.
- Customers within each cluster are more similar before merging with other clusters.
- The larger vertical gaps indicate good separation between customer groups.
- Around 3 clusters can be identified by cutting the dendrogram at a suitable distance.
- Complete Linkage produces more compact and balanced clusters, making it more suitable than Single Linkage for this dataset.

c. Average Linkage



- The dendrogram shows clear and reasonably well-separated clusters.
- Customers within each cluster are fairly similar before merging with other groups.
- Around 3 clusters can be identified by cutting the dendrogram at an appropriate distance.
- Average Linkage produces balanced clusters and reduces the chaining effect seen in Single Linkage.
- It provides better clustering than Single Linkage.

d. Ward Linkage



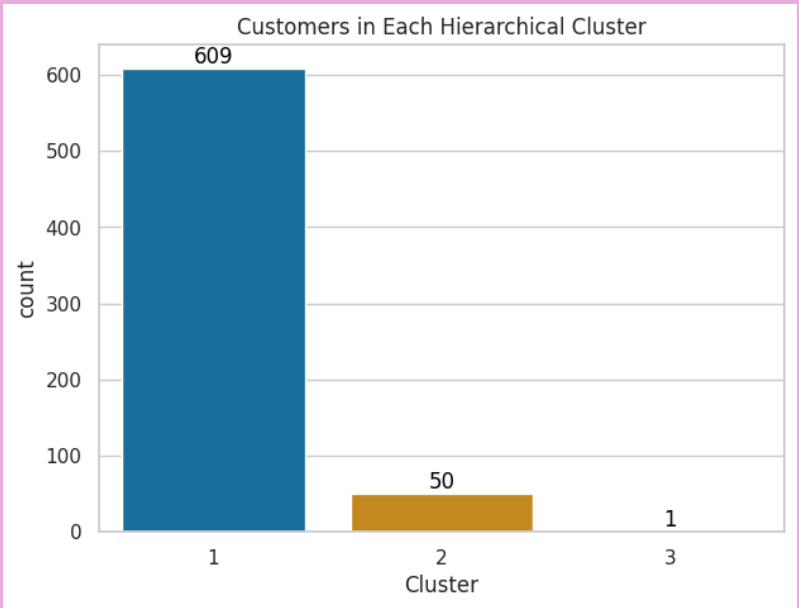
- The dendrogram shows clear and well-separated customer clusters.
- Customers within each cluster are highly similar, resulting in compact groups.
- A cut at an appropriate distance suggests 3 distinct clusters.
- Ward Linkage minimizes the variation within each cluster, producing balanced and compact clusters.
- Ward Linkage is the most suitable linkage method for this dataset because it provides the clearest cluster separation and the best clustering performance.

➢ Cophenetic Correlation

Linkage Method	Cophenetic Correlation	Interpretation
Average	0.880	Best fit to the data
Complete	0.844	Very good clustering
Ward	0.764	Good clustering
Single	0.682	Least suitable

➢ Final Hierarchical Clustering Model

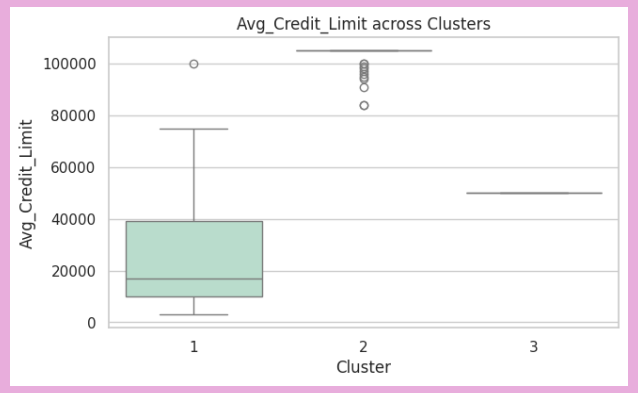
	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Customer_Engagement	Digital_Ratio	Cluster
0	100000.0	2.0	1.0	1.0	0.0	2.0	0.5	1
1	50000.0	3.0	0.0	8.5	9.0	17.5	8.5	3
2	50000.0	7.0	1.0	3.0	4.0	8.0	1.5	1
3	30000.0	5.0	1.0	1.0	4.0	6.0	0.5	1
4	100000.0	6.0	0.0	8.5	3.0	11.5	8.5	2



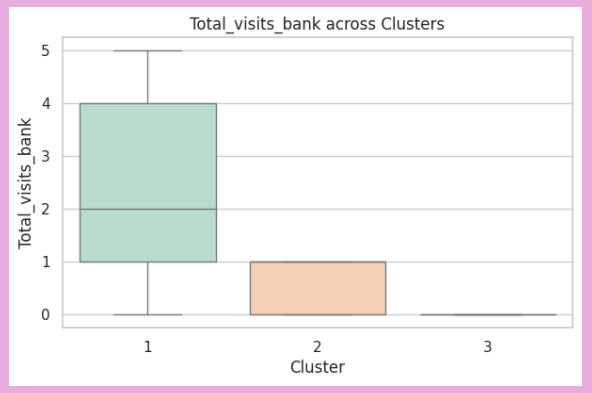
- Cluster 1 contains 609 customers, making it the largest cluster (about 92% of customers).
- Cluster 2 contains 50 customers, representing a small customer segment (about 7.6%).
- Cluster 3 contains only 1 customer, indicating an outlier or unique customer.
- The clusters are highly imbalanced, with most customers grouped into a single cluster.

➢ Cluster Profiling

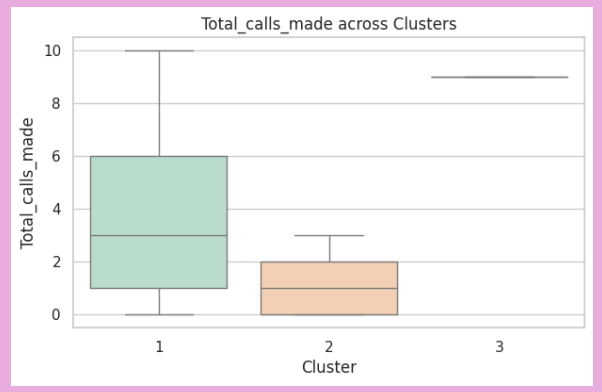
Cluster	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Customer_Engagement	Digital_Ratio
1	25807.881773	4.377668	2.555008	1.912972	3.779967	8.247947	0.951095
2	102660.000000	8.740000	0.600000	8.180000	1.080000	9.860000	5.745000
3	50000.000000	3.000000	0.000000	8.500000	9.000000	17.500000	8.500000

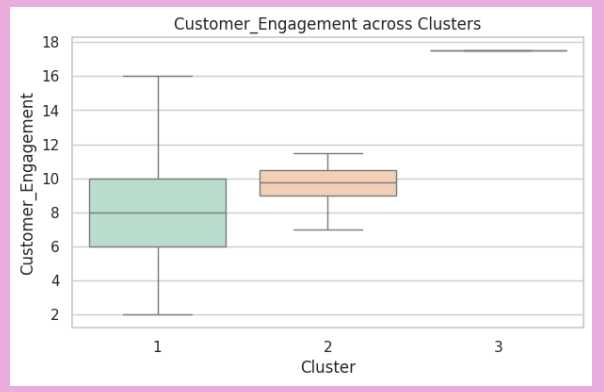


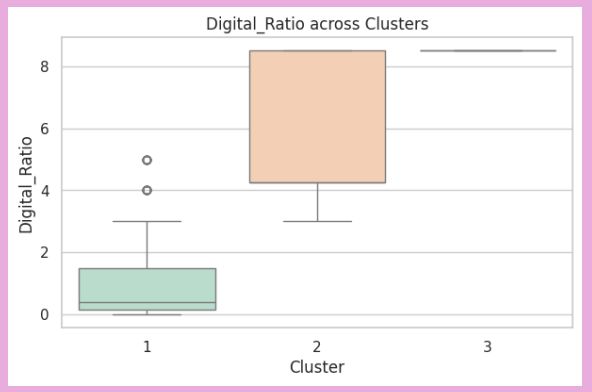




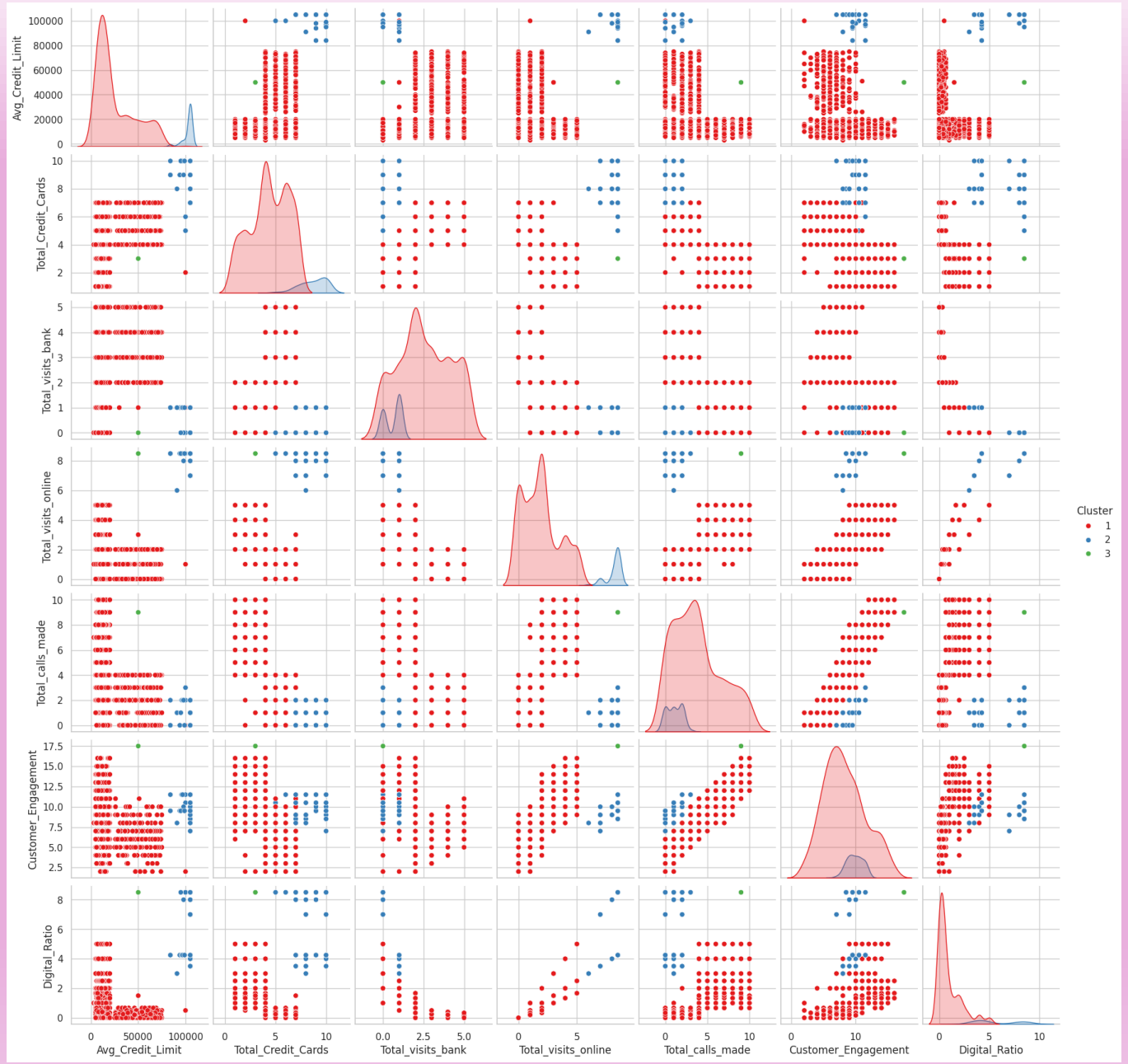






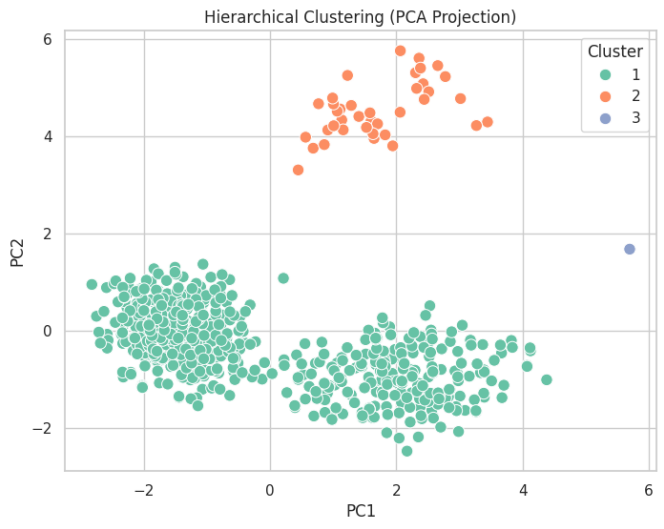


➢ Pairplot of Clusters Profiling



- Cluster 1 contains the majority of customers with moderate credit limits and average banking behavior.
- Cluster 2 represents high-value customers with the highest credit limits, more credit cards, and higher online activity.
- Cluster 3 contains only one customer, making it an outlier with unique characteristics.
- Customers with higher credit limits generally own more credit cards and make fewer customer service calls.
- The pair plot shows less balanced clusters compared to K-Means, with one dominant cluster and one outlier cluster.

➢ PCA Visualization of Clusters Profiling



- The PCA plot clearly shows three distinct customer clusters, indicating successful segmentation by the K-Means algorithm.
- Cluster 1 is well separated from the other clusters, representing customers with unique characteristics.
- Clusters 0 and 2 are also clearly distinguishable with very little overlap.
- Most customers are concentrated in Clusters 0 and 2, while Cluster 1 contains fewer customers.
- The clear separation between clusters suggests that the selected K = 3 provides meaningful and well-defined customer segments.

➢ Cluster Summary Table

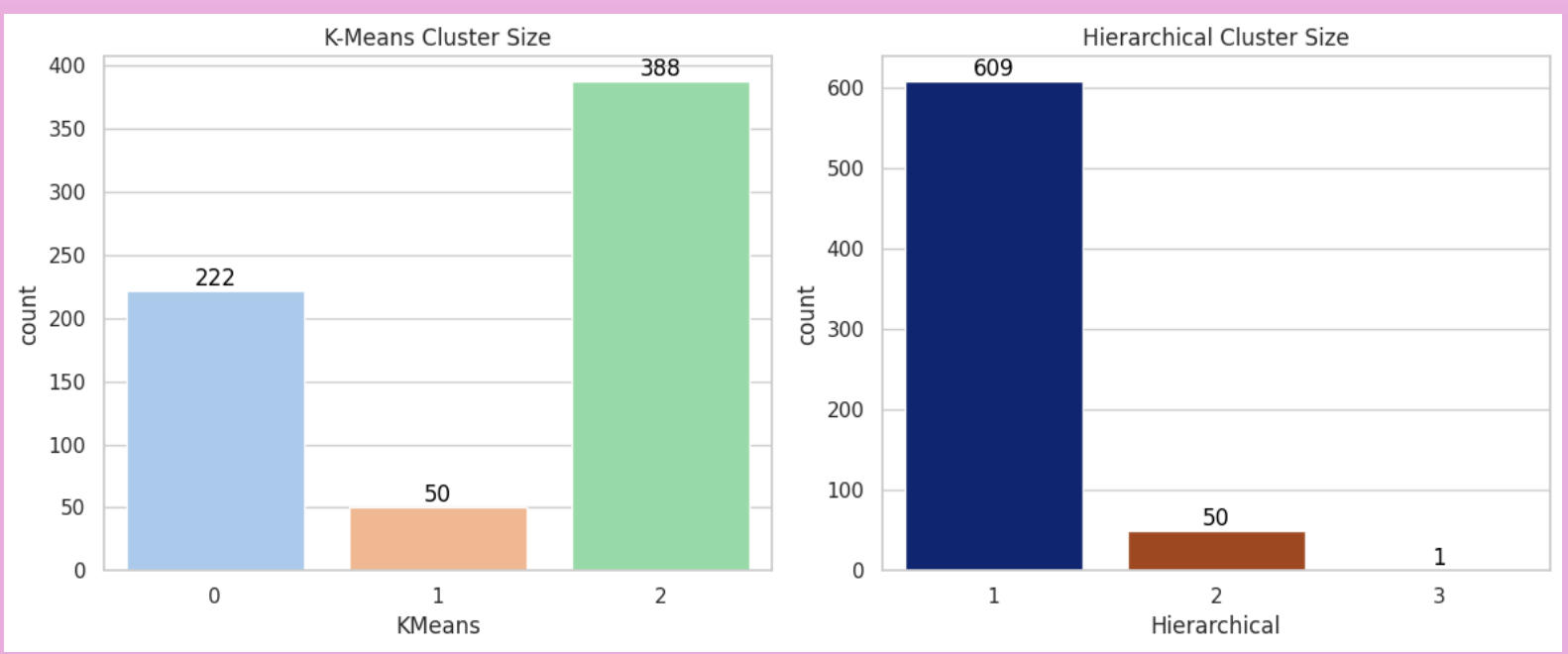
Cluster	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
1	25807.88	4.38	2.56	1.91	3.78
2	102660.00	8.74	0.60	8.18	1.08
3	50000.00	3.00	0.00	8.50	9.00

➢ Final Conclusion

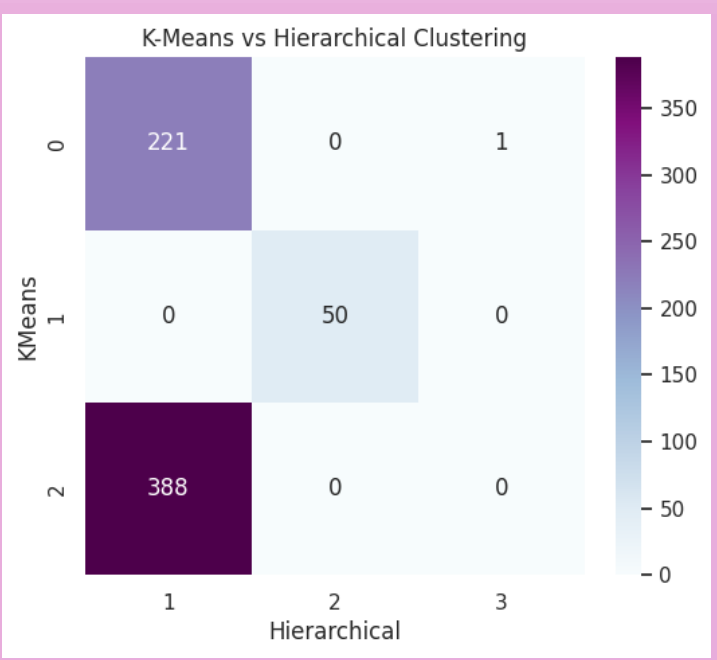
- Hierarchical Clustering identified 3 customer clusters based on banking behavior and spending patterns.
- Average Linkage performed the best with the highest cophenetic correlation (0.88), followed by Complete Linkage (0.844).
- The clustering results were imbalanced, with one large cluster, one small cluster, and one outlier customer.
- Compared to K-Means, Hierarchical Clustering produced less balanced customer segments.

6. K-means vs Hierarchical Clustering

➢Compare Cluster Sizes



➢Compare Cluster Sizes



➢ Compare Cluster Profiling

- **K-Means Cluster Profile**

Cluster	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Customer_Engagement	Digital_Ratio
0	12184.684685	2.400901	0.932432	3.565315	6.909910	11.407658	2.234985
1	102660.000000	8.740000	0.600000	8.180000	1.080000	9.860000	5.745000
2	33664.948454	5.505155	3.476804	0.984536	2.002577	6.463918	0.235954

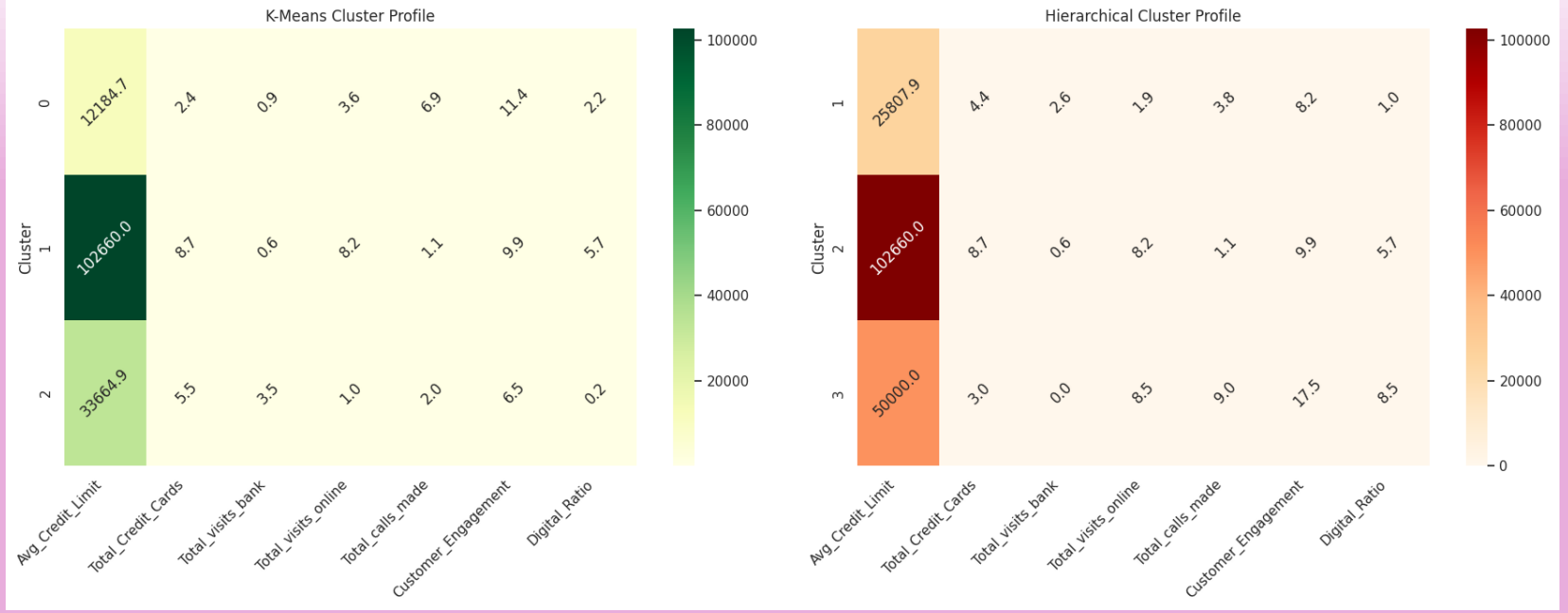
- **Hierarchical Cluster Profile**

Cluster	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Customer_Engagement	Digital_Ratio
1	25807.881773	4.377668	2.555008	1.912972	3.779967	8.247947	0.951095
2	102660.000000	8.740000	0.600000	8.180000	1.080000	9.860000	5.745000
3	50000.000000	3.000000	0.000000	8.500000	9.000000	17.500000	8.500000

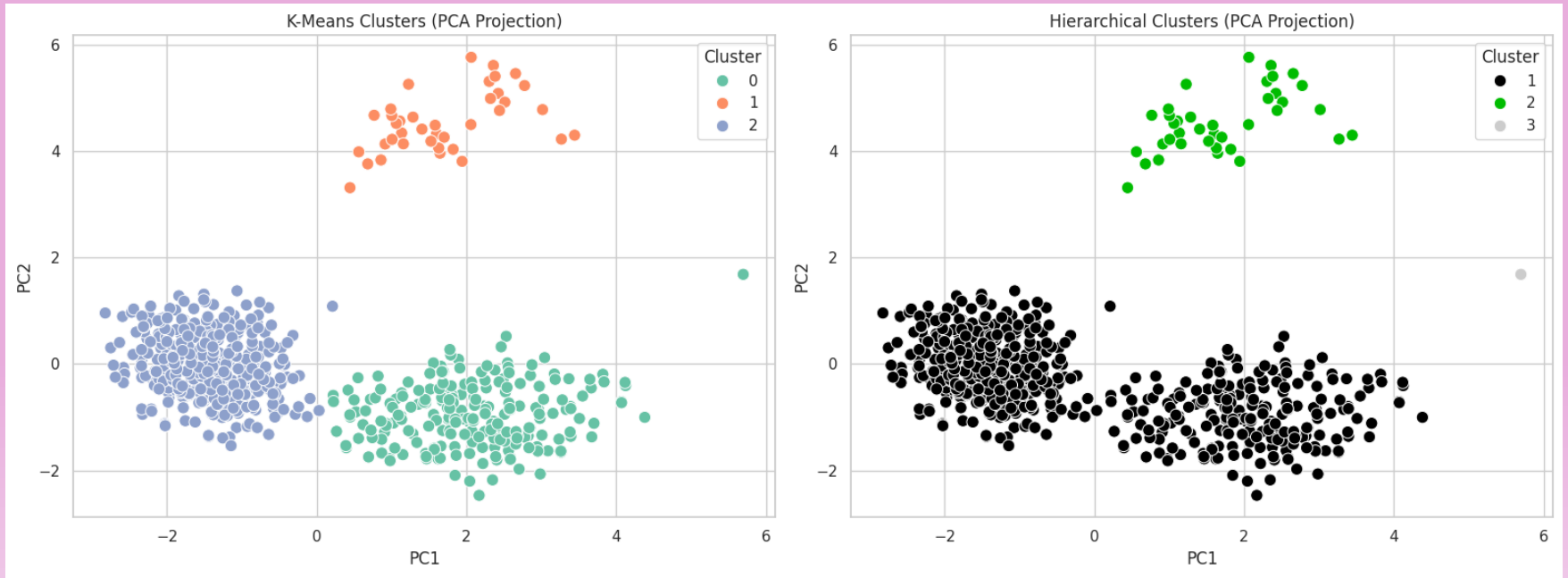
➢ Silhouette Score Comparison

- K-Means Silhouette Score : 0.508
- Hierarchical Silhouette Score : 0.463

➢ Heatmaps Comparison



➢ PCA Comparison



7. Actionable Insights & Recommendations

1. High-Value Customers Form a Distinct Segment

Insight

- A customer segment has significantly higher average credit limits and owns multiple credit cards.
- These customers primarily use digital banking channels and rarely visit bank branches or contact customer support.
- They represent the bank's most profitable customer group.

Recommendation

- Launch premium credit card upgrade campaigns.
- Offer exclusive reward programs, airport lounge access, cashback, and travel benefits.
- Assign dedicated relationship managers for high-net-worth customers.
- Provide personalized investment and wealth management products.

2. Branch-Oriented Customers Need Digital Adoption

Insight

- A significant customer segment frequently visits bank branches while making relatively few online visits.
- This indicates dependence on traditional banking services.

Recommendation

- Promote digital banking through awareness campaigns.
- Offer incentives such as cashback or reward points for using mobile and internet banking.
- Conduct branch-based demonstrations on digital banking features.
- Simplify the digital onboarding process.

3. Customers with Frequent Service Calls Need Better Support

Insight

- Some customers make a high number of customer service calls despite having lower credit limits.
- Frequent service interactions may indicate dissatisfaction or unresolved issues.

Recommendation

- Improve first-call resolution rates.
- Introduce AI-powered chatbots and self-service portals for common queries.
- Provide proactive notifications for billing, transactions, and card status.
- Analyze call reasons to identify recurring service issues.

4. Customers with Low Credit Limits Have Growth Potential

Insight

- A large portion of customers have relatively low credit limits and own only one or two credit cards.
- These customers represent an opportunity for future growth.

Recommendation

- Offer credit limit enhancement programs based on repayment behavior.
- Cross-sell additional financial products such as loans, insurance, and savings accounts.
- Introduce entry-level reward credit cards with attractive benefits.

5. Digital Customers Are More Self-Sufficient

Insight

- Customers who frequently use online banking tend to visit branches and call customer support less often.
- Digital engagement is associated with lower servicing costs.

Recommendation

- Encourage all customer segments to adopt digital channels.
- Continuously improve the mobile banking application and internet banking platform.
- Offer personalized recommendations through digital channels.

6. Customer Behavior Is Heterogeneous

Insight

- Clustering analysis confirms that customers differ significantly in terms of spending behavior, banking preferences, and service interactions.
- A single marketing strategy is unlikely to be effective across all customers.

Recommendation

- Develop segment-specific marketing campaigns.
- Personalize product recommendations based on customer profiles.
- Use customer segmentation as an ongoing strategy for campaign planning.

Overall Business Recommendations

- Implement customer segmentation as part of the bank's CRM strategy.
- Personalize marketing campaigns based on customer cluster profiles.
- Upsell premium credit cards to high-value customers.
- Increase digital banking adoption through targeted initiatives.
- Improve customer support efficiency by addressing common service issues.
- Regularly retrain the clustering model as new customer data becomes available.
- Monitor key performance indicators (KPIs) such as customer retention, campaign conversion, digital adoption, and customer satisfaction to measure the effectiveness of segmentation strategies.

Final Conclusion

- The clustering analysis demonstrates that AllLife Bank's customers can be segmented into three distinct groups, each with unique financial behavior and service preferences. Both K-Means and Hierarchical Clustering identified similar customer profiles, validating the robustness of the segmentation.
- Among the two techniques, K-Means is recommended as the preferred model due to its computational efficiency, scalability, and ease of deployment. By leveraging these customer segments, AllLife Bank can deliver personalized marketing campaigns, improve customer support, encourage digital banking adoption, and optimize resource allocation. These data-driven strategies are expected to enhance customer satisfaction, improve operational efficiency, increase cross-selling opportunities, and strengthen the bank's overall profitability.